

Component X Net-Near Forging — Quench & BFDF

Slide notes. Audience: mixed (eng + management). Conceptual, no equations. Values: 1550 °F soak → 75 °F water quench.

Slide 1 — Component X Net-Near Forging

On slide

- **Net-near forging:** one forged piece replaces two welded forgings → saves material + machining, **eliminates a major weld** across the cylindrical face
- **Quench (+ temper):** rapid cool from ~1550 °F to give a heavy forging the strength *and* toughness it needs
- **Charpy test:** impact specimen cut from the *actual* forging → fracture toughness at the limiting location
- **Path:** Abaqus quench sim → RTT_quench → BFDF margin vs. design factor

Talking points

- This is a tech-development drawing — we're checking whether a vendor can forge Component X as a single net-near piece.
 - Today it's two forgings welded across the cylindrical face. Going to one forging drops that weld, saves material, cuts cost.
 - Trade-off: a thicker single forging cools slower in the quench, and slower cooling means lower toughness in the interior.
 - So I ran a quench analysis to confirm we keep enough fracture margin — the sim gives RTT_quench, and RTT_quench sets the BFDF margin.
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Slide 2 — Analysis Setup

Visual

- 2D axisymmetric cross-section of the forging
 - Outer (wetted) surface highlighted; arrow → box "**film coefficient, $h(T)$** " (sink temp $T_\infty = 75$ °F)
 - Box inside the section: "**1550 °F**" (initial soak)
- Animation: temperature vs. time — starts 1550 °F, cools toward 75 °F water
 - Arrow → box "**Charpy location**" at the specimen cut location

On slide

- Axisymmetric 2D model — cross-section captures the full part
- Starts 1550 °F, quenched into 75 °F water; convective cooling $h(T)$ on wetted surfaces

- Interior lags the surface — locate the slowest-cooling point (longest time-to-cool)
- Take RTT_quench at that worst-case point → bounds the part regardless of flaw location

Talking points

- Model's 2D axisymmetric — the part's a body of revolution, so one cross-section captures the whole thing.
 - Starts hot at 1550, dumped into 75° water. Surfaces cool in seconds; the interior takes far longer — that's the animation.
 - I care about the interior: the slowest-cooling spot has the coarsest microstructure and the worst toughness.
 - I pull RTT_quench at that worst-case location, so it bounds the forging no matter where the flaw actually sits.
 - Then I compare it to where the Charpy specimen is cut — that gap is what makes the number conservative.
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Slide 3 — BFDF Margin Results

Alt titles: "Result: Margin Confirmed" · "Fracture Toughness Impact"

Visual

- Screenshot: BFDF workbook — margins for each cutline across the reports where Component X is used
- Chart — BFDF per cutline, three series **Base** / **Min material** / **Max material**, with a horizontal **DF-limit** line; all series above it
 - *(Axis note: bars = BFDF ratio, line = DF value. If you'd rather plot margin (BFDF – DF), the limit line sits at 0.)*

On slide

- BFDF evaluated for every cutline / report using Component X
- Min & max material properties bracket the result
- All cases clear the design factor — BFDF stays above the DF limit (> 2)

Talking points

- Bottom line: BFDF clears the design factor for every cutline where this part shows up.
- The three bars per case bracket material scatter — min and max properties — and even the low side stays above the DF line.
- Margin's comfortable, above 2 across the board.
- That's enough for the design group to sign off on the drawing.

- So we release the tech-development drawing to the vendor and see if they can forge it as one piece.